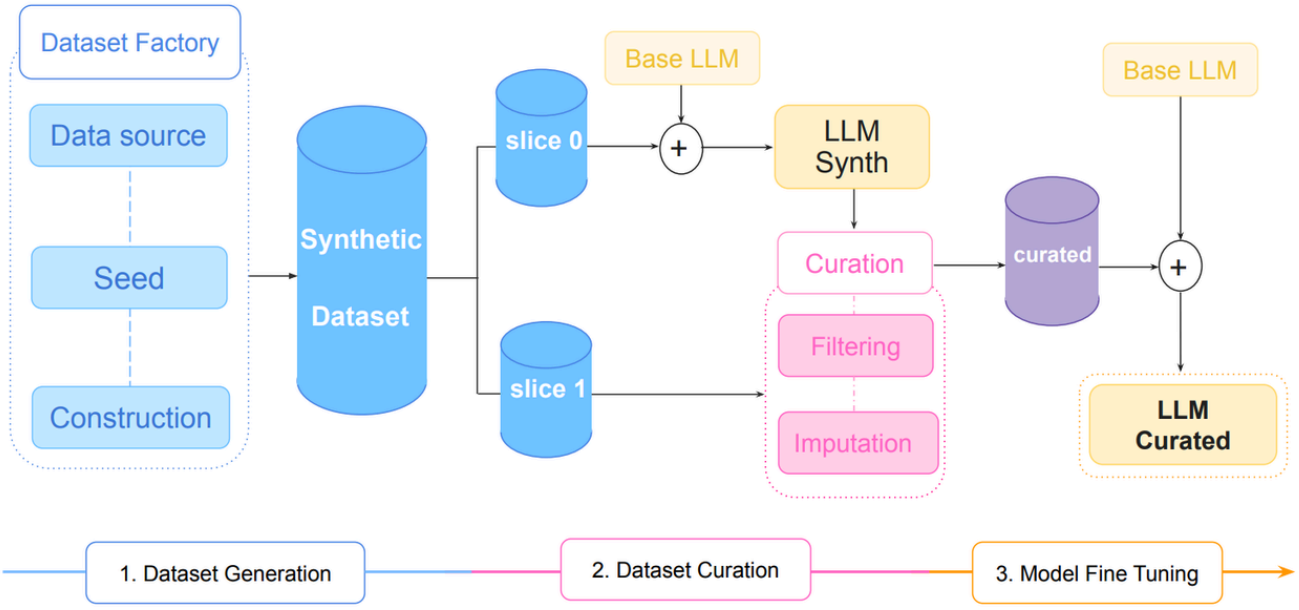
Source2Synth: Synthetic Data Generation and Curation Grounded in Real Data Sources

> #Dataset #Finetuning



- What problems does this framework solve?
  - When handling multi-step inference or tool calling, we need a lot of high-quality data to finetune the model, which is hard to find.
  - It is not feasible to label the large amount of data; letting the LLM generate is not reliable either, as the hallucination will pollute the dataset.
  - This framework will **anchor** the data generation to Real-world Data Sources to generate high quality data without human interference.
- What does this framework do?
  - This framework generates data in three steps to generate two types of data: multi-hop QA (MHQA) and tabular QA (TQA).
  - Step 1: Data Generation
    - Data Source Selection:
      - MHQA: Use English Wiki. Randomly select a document D_1, and select one related document of D_1, denoted as D_2.
      - TQA: Use non-labeled tables from WikiSQL.
    - Seed: To ensure the relationship of the datasets, we select a "seed"/"anchor".
      - MHQA: Select an entity that is both in D_1 and D_2, serving as a "hop" connection.
      - TQA: Use a model to extract an "interesting fact" from the table, serving as a SQL question.
    - Dataset construction: Simulate the logic of CoT to generate the dataset.
      - MHQA: Generate a question Q_1 based on D_1 such that the **answer to it** is E.

Generate another question Q_2 based on D_2 such that its **topic** is E. Merge Q_1, Q_2 to form Q.
        - TQA: With zero-shot, generate a SQL query based on the table and the "interesting fact" (question). Test the query on the table; discard it if it cannot be executed properly.
    - Dataset Curation
        - Data filtering: The generated synthetic dataset is divided into two equal halves (Slice 0 and Slice 1). First, Slice 0 is used to fine-tune an intermediate language model, denoted as LLM_{Synth}. Subsequently, LLM_{Synth} is employed to clean the data in Slice 1, comprising the following two steps:
            1. Filtering: Let LLM_{Synth} run k times on Slice 1. If it cannot answer the question, then the sample is discarded.
            2. Data imputation (*for MHQA only*): Given Q, Q_2, the entity and D_1, let the LLM recover the Q_1 (Q_1'). If Q_1' and Q_2 can conduct the same answer as the original Q_1 and Q_2 can, then keep the sample.
    - Model fine-tuning (SFT).
- Experiment Results:
    - Significant performance improvement: On the WikiSQL (TQA) task, the Source2Synth method achieved a 25.51% performance boost compared to the fine-tuned baseline model. On the HotpotQA (MHQA) task, performance improved by 22.57%.
    - Data scaling law proves effective: Experiments demonstrate that even with minimal data, model performance steadily improves by adding more clean synthetic data to the fine-tuning data pool.
    - The necessity of anchoring to real-world data sources: The paper compared synthetic data generated from “completely ungrounded” vocabulary and found that data anchored to real-world sources like Wikipedia not only significantly reduced text perplexity (PPL) from 24.7 to 13.6 but also delivered an absolute accuracy gain of approximately 7%.
- Work Cited
    - Paper
    - NotebookLM
    - Chat History